

Criterion A: Planning

Description of Scenario:

Madalena is a grade 12 student at my school who founded and is currently managing “Students for Students”. An initial interview was conducted at our school on the 22nd of May 2024(Appendix A). The tutoring organization connects students who need academic support with student tutors from our school. They get requests from people who need tutoring and students who want to tutor, and they then match up. I chose them as a client due to my past experience as one of their tutors.

The organization currently has its own Google Sites website to attract tutors and students and makes them fill out a Google Form with their details and personal preferences, and this data is collected in a Google Sheet for each subject. They highlight the names in different colors to indicate that they have been interviewed or matched up. After matchups, occasional Google Forms are sent to collect feedback.

The current system for Students for Students is not sustainable in the long run for a few reasons. Firstly the matchup process is very time-consuming “I don't have the time for manual matching anymore ”(see Appendix A). Madalena has to manually select the matchups and send the emails. Also, this information is spread across a large spreadsheet making it difficult to maintain an overview of the tutor and student. These problems could potentially force Students for Students to shut down or compromise on their organization’s 48-hour response.

Word count: 233

Rationale for the Proposed Solutions:

Before approaching Madalena about my IA, I suggested to her that she could enlist more people to help her manage the student/tutor database. She said that adding more people makes it harder for admins to coordinate but she would eventually be forced to do that unless there is another way to maintain the response time per request. When I suggested a program that would allow admins to match students and tutors, she was very interested because it allowed her to maintain control without having to sacrifice the organization’s 48-hour response time policy which she nowadays struggles to fulfill stating “I get angry emails from either the parents or the students who have been waiting for so long”(see Appendix A)

I suggested 2 other options to the client: a web-based application using PHP and a desktop application using Java. However, we concluded that developing an Android Application would be the best solution since the application would be accessible to her at all times and she could match up students on the go. An Android App (.apk) can work on her Samsung Galaxy S24 FE and once she graduates it she can transfer the app to Vice President Luisa’s Galaxy A55 5G. Initially, I wanted to use Android Studio, but I decided to use the MIT App Inventor instead

because of its ability to provide a visual, block-based programming environment to which Madalena can provide regular feedback as well as its ability to integrate with Google Sheets using Web API.

Word count: 248

Success Criteria:

- Is able to authenticate app users through login credentials stored/managed by the client in an external database
- Matches students correctly with suitable tutors using a compatibility algorithm with at least a 90% success rate
- Reduces the time my client takes to match students with tutors to under 1,5 min
- Imports student/tutor/login data from Google Sheets
- Updates Google Sheets with unique matchup ID once students get matched up with a tutor
- Displays a list of all current unmatched students that will be matched
- Displays lists of all current matched tutors, matched students, and unmatched tutors
- Has an easily navigable graphical user interface (based on client feedback)
- Includes error handling to manage invalid data entry from the client
- Can send bulk emails to all tutors or students